ERA 5 calibration to Elexon power

S. Gomez[1]

[1] School of Mathematics, University of Edinburgh

2026-07-09

Table of contents I

Updates

New

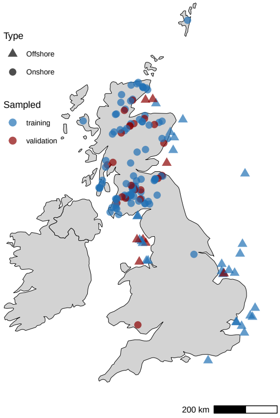
- OOS assessment
  - OOS time
  - OOS locations
- Low wind events distribution

Model validation framework

Framework for OOS assessment

- Run models on a sample of days
  - 80/20 WF split for training and testing
  - Training with 3 days to predict the 4th
  - Randomly chosen 15 days from 2025, 5 per regime
- Evaluate performance at WF level
- Error under aggregation
  - GB level
  - Onshore / offshore

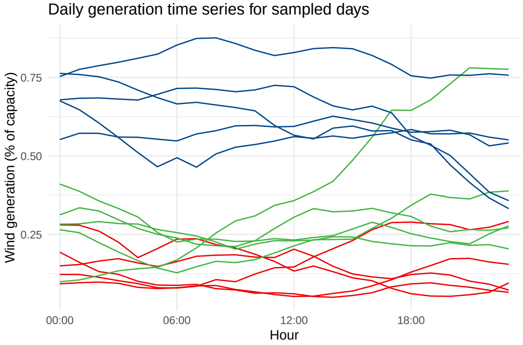
Sampled days for performance assessment



Sampled days in different regimes

- Objective is to assess performance by regime
- OOS days are different regimes of wind speed
- Previous 3 days can be anything

Sampled days for performance assessment

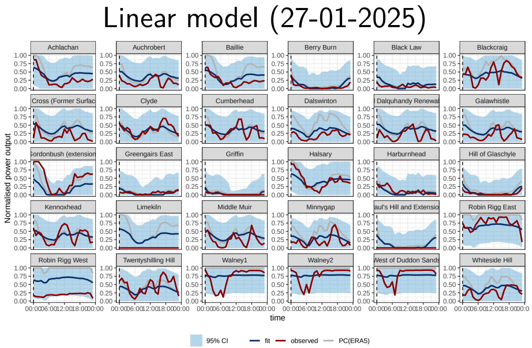


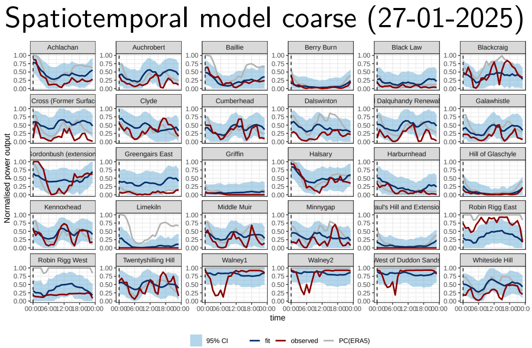
Model portfolio

- Wind farm level models:
  - Generic Power Curve (ERA5 + generic PC estimate)
  - Quantile Mapping
  - Linear model
  - AR1, AR2
  - Spatiotemporal model fine mesh and coarser mesh
- Linear model at Great Britain (GB) level

Out-of-sample time

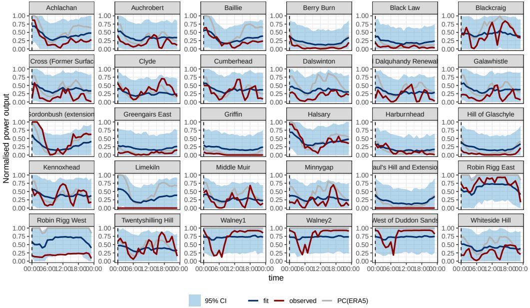
Wind farm prediction bands



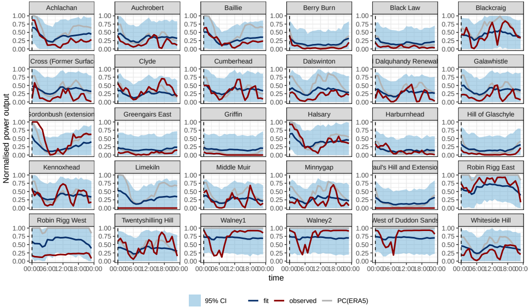


Wind farm prediction bands

1D SPDE model (27-01-2025)

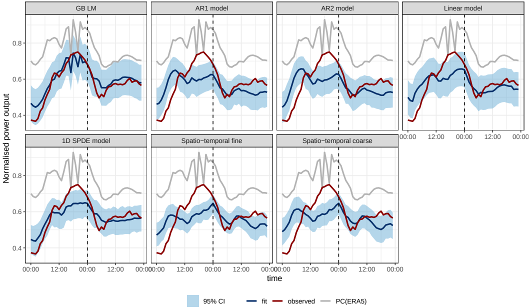


AR1 model (27-01-2025)

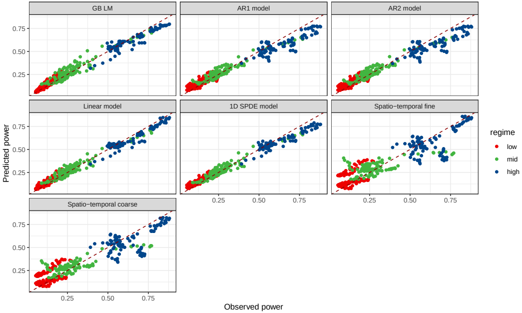


Aggregated prediction bands

GB level



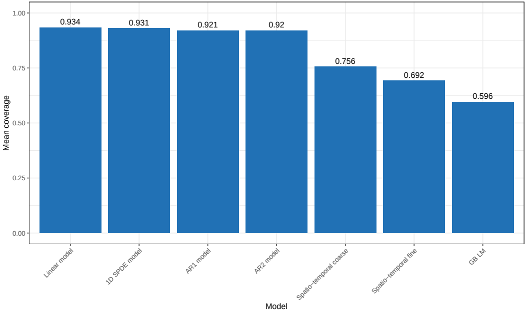
Scatter plot oos



Day-ahead forecast coverage

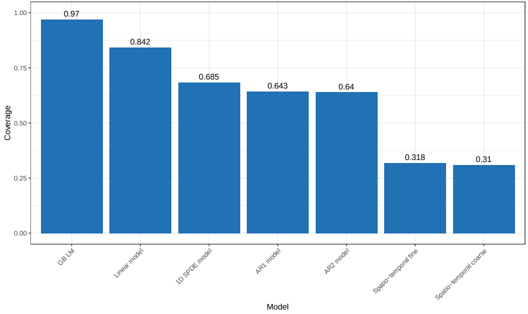
- Models at WF can achieve target coverage
- Aggregation reduces coverage
- High regime more succeptible to coverage loss
- Spatiotemporal model is not improving coverage

Coverage for day-ahead predictions

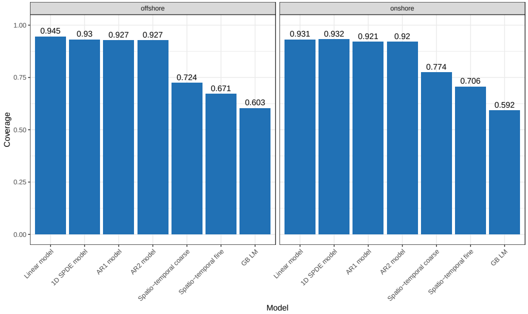


Coverage under aggregation

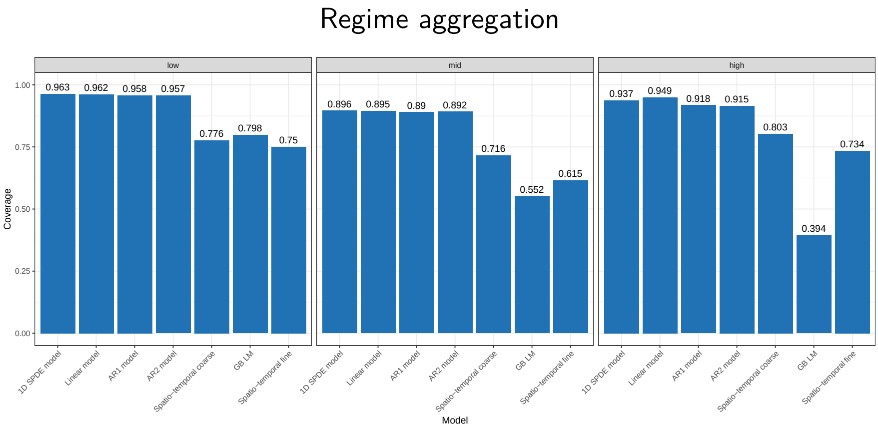
GB level



Onshore / offshore



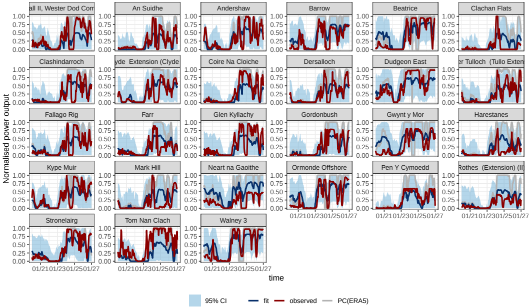
Coverage under aggregation by regime



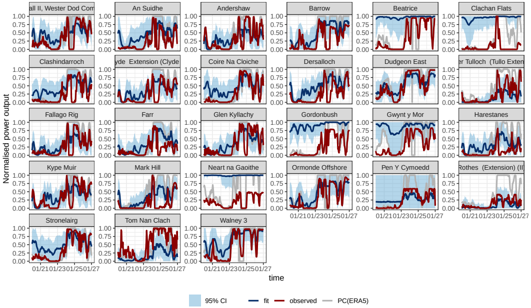
Out-of-sample locations

Wind farm prediction bands

Linear model (27-01-2025)

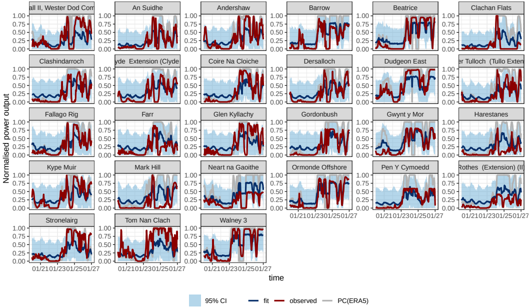


Spatiotemporal model coarse (27-01-2025)

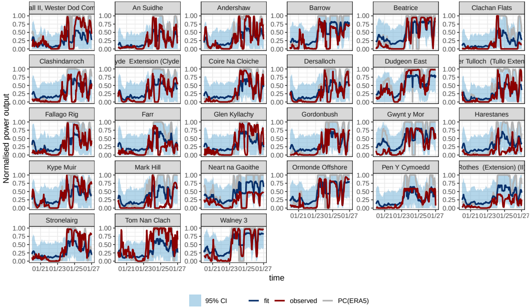


Wind farm prediction bands

1D SPDE model (27-01-2025)

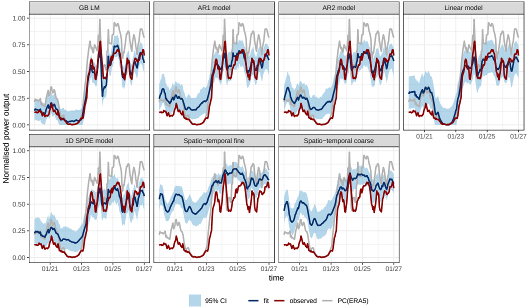


AR1 model (27-01-2025)

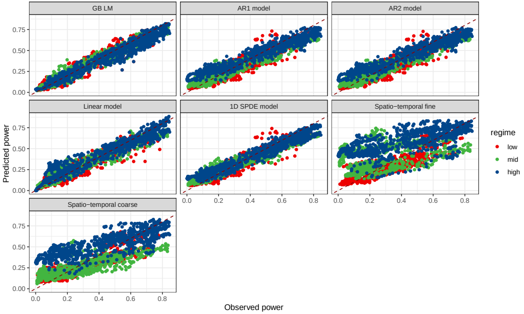


Aggregated prediction bands

GB level (27-01-2025)

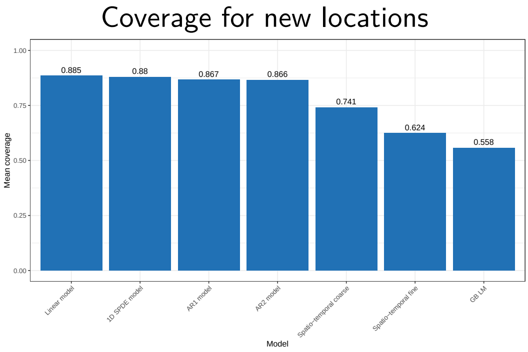


Scatter plot oos



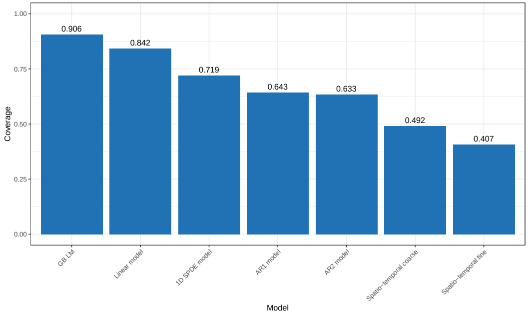
Unseen locations prediction intervals coverage

- Attaining coverage in unseen locations is more challenging
- Isolated locations can lead to high uncertainty
- Location specific events may affect coverage
- Spatiotemporal model is not improving coverage

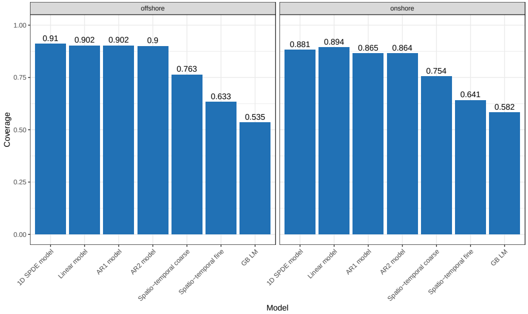


Coverage under aggregation

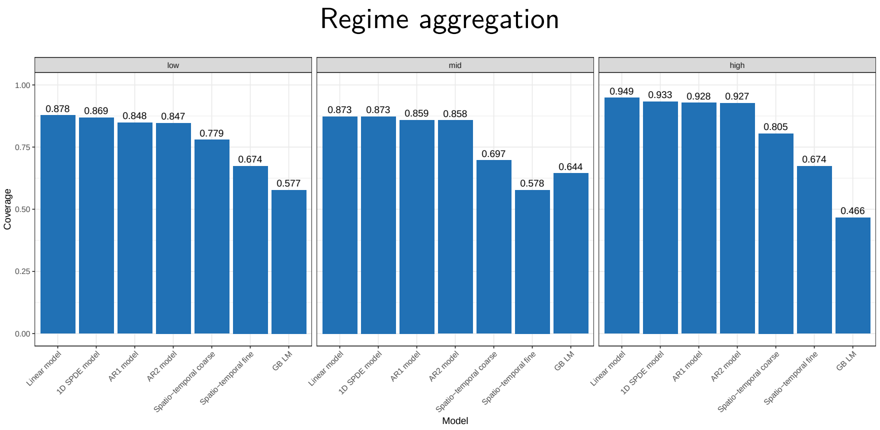
GB level



Onshore / offshore



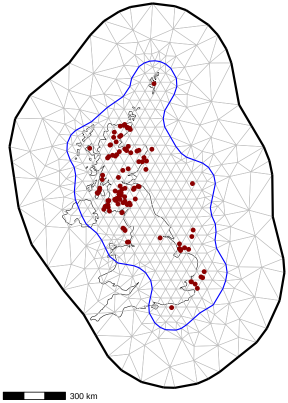
Coverage under aggregation by regime



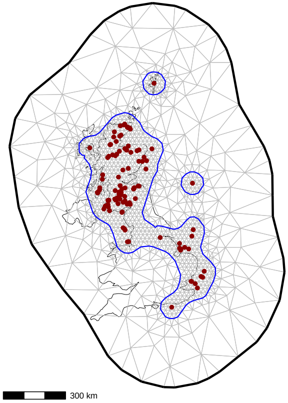
Appendix

Meshes for spatiotemporal models

Coarse mesh



Fine mesh



Error metrics and performance assessment

Error metrics

model	RMSE	MAE	Bias
QM	0.210	0.135	1.58e-04
GB LM	0.191	0.130	2.12e-02
Linear model	0.168	0.117	1.76e-03
1D SPDE model	0.167	0.116	1.46e-03
AR2 model	0.164	0.113	1.45e-03
AR1 model	0.164	0.113	1.44e-03
Spatio-temporal coarse	0.124	0.082	1.26e-03
Spatio-temporal fine	0.097	0.061	1.16e-03

Error metrics by regime

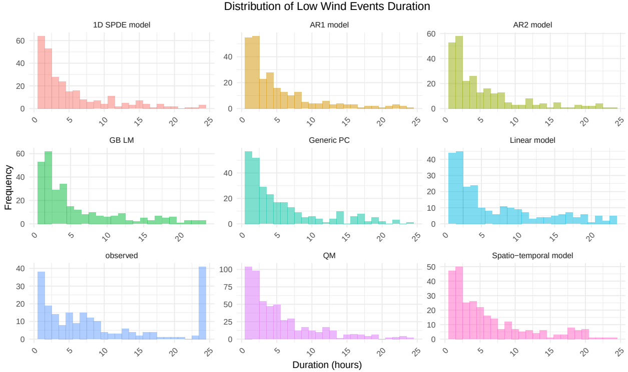
	Low			Mid			High		
Model	RMSE	Bias	MAE	RMSE	Bias	MAE	RMSE	Bias	MAE
QM	0.110	0.062	-1.59e-03	0.196	0.133	1.08e-04	0.305	0.219	2.21e-03
GB LM	0.108	0.069	2.05e-02	0.186	0.132	2.44e-02	0.262	0.193	1.44e-02
1D SPDE model	0.093	0.059	7.34e-03	0.160	0.115	2.50e-03	0.234	0.179	-7.56e-03
Linear model	0.092	0.059	6.57e-03	0.162	0.117	2.26e-03	0.235	0.183	-4.75e-03
AR2 model	0.089	0.056	3.07e-03	0.156	0.112	1.24e-03	0.230	0.176	1.49e-04
AR1 model	0.089	0.056	3.05e-03	0.156	0.112	1.22e-03	0.230	0.176	1.85e-04
Spatio-temporal coarse	0.061	0.037	2.22e-03	0.117	0.081	1.13e-03	0.180	0.132	5.07e-04
Spatio-temporal fine	0.048	0.028	1.75e-03	0.088	0.059	1.05e-03	0.145	0.101	7.84e-04

Error metrics by technology

	Offshore			Onshore		
Model	RMSE	Bias	MAE	RMSE	Bias	MAE
QM	0.213	0.139	5.85e-02	0.209	0.133	-1.85e-02
GB LM	0.184	0.127	8.26e-03	0.193	0.131	2.54e-02
1D SPDE model	0.172	0.119	6.88e-04	0.166	0.114	1.70e-03
AR1 model	0.169	0.117	6.72e-04	0.162	0.111	1.68e-03
AR2 model	0.169	0.117	7.36e-04	0.162	0.111	1.67e-03
Linear model	0.157	0.113	2.84e-03	0.172	0.118	1.42e-03
Spatio-temporal coarse	0.098	0.066	1.26e-03	0.131	0.087	1.26e-03
Spatio-temporal fine	0.073	0.048	5.82e-04	0.103	0.065	1.34e-03

Low wind speed events

Low wind speed events distribution



Spatiotemporal calibration model

Model overview

$$\mathbb{E}(p_{it}^{\text{calib}}) = \alpha + \beta_k + \beta_m + \beta_h + s_p(\hat{p}_{it}) + s_w(w_{it}) + u_{it}, \tag{1}$$

- Calibrated power: corrected output at site *i*, time *t*.
- α: intercept.
- Fixed effects $\beta_k, \beta_m, \beta_h$: location/season/time structure.
- ERA5 correction s_p: bias adjustment of ERA5 + generic PC estimate.
- Wind effect s_w: nonlinear wind–power correction.
- Spatiotemporal effect u_{it}: Spatial component + AR1 error.

References

Cannon, D. J., D. J. Brayshaw, J. Methven, P. J. Coker, and D. Lenaghan. 2015. "Using Reanalysis Data to Quantify Extreme Wind Power Generation Statistics: A 33 Year Case Study in Great Britain." *Renewable Energy* 75 (March): 767–78. https://centaur.reading.ac.uk/38448/.

Potisomporn, P. 2024. "Spatiotemporal Variability of Extreme Low Wind Power Events in Great Britain - ORA - Oxford University Research Archive." Http://purl.org/dc/dcmitype/Text. https://ora.ox.ac.uk/objects/uuid:de37bee2-1813-42e5-a69c-9b15f038d096.

Thornton, Hazel E. 2020. "Atmospheric Circulation, Seasonal Predictability and Britain's Energy System." PhD thesis, University of Reading. https://doi.org/10.48683/1926.00095832.